Contents

1 Appendix B: Environmental Channel Modeling **2**

1.1 Objective . . . 2

1.2 Methods (src) . . . 2

1.3 Script and outputs . . . 2

1.4 Figure . . . 2

1.5 Equation references . . . 2

1.6 Reproducibility . . . 2

1.7 Context Note on Biological Ranges . . . 2

1.8 Cross-references . . . 3

1 Appendix B: Environmental Channel Modeling

1.1 Objective

Comprehensive atmospheric channel modeling validated against peer-reviewed atmospheric physics: molecular absorption (H_2O, CO_2, CH_4, O_3), Rayleigh scattering, aerosol effects, channel capacity mapping with 8-14 μm window emphasis, wavelength optimization for 10-100 m ranges, and environmental sensitivity analysis for IR communication in insect olfactory systems.

1.2 Methods (src)

- `src/case_studies/environmental_channel.py`
  - `molecular_absorption_cross_section(wavelengths, molecule_type)` — H2O, CO2, CH4 absorption
  - `rayleigh_scattering_coefficient(wavelengths, air_density)` — molecular scattering
  - `atmospheric_transmission_comprehensive(wavelengths, conditions)` — multi-component transmission
  - `channel_capacity_analysis(wavelengths, environmental_conditions)` — Shannon capacity mapping
  - `optimize_wavelength_for_range(target_range, capacity_requirements)` — wavelength selection
  - `environmental_sensitivity_analysis(parameter_variations)` — parameter sensitivity
  - `atmospheric_transmission_detailed(wavelengths, humidity, temperature, path)` — basic transmission utility
  - `channel_capacity_vs_env(material_props, env_grid)` — grid mapping of capacity vs environment

1.3 Script and outputs

- Script: `scripts/generate_environmental_channel_analysis.py`
- Data: `output/data/environmental_channel_comprehensive.npz`
- Figure: `figures/environmental_channel_comprehensive_analysis.png`

1.4 Figure

1.5 Equation references

- Atmospheric transmission: see (??)
- Channel capacity: see (??)

1.6 Reproducibility

- Run: `python3 scripts/generate_environmental_channel_analysis.py`
- Artifacts saved to `output/data/` and `figures/`.
- Deterministic grids via `src/config.set_random_seed(42)`.

1.7 Context Note on Biological Ranges

Some insects exhibit sensitivity to thermal IR in natural behaviors (e.g., Aedes aegypti up to ~0.7 m; Chandel et al. 2024). These thermal constraints complement our electromagnetic window analysis and motivate species- and wavelength-specific range predictions.

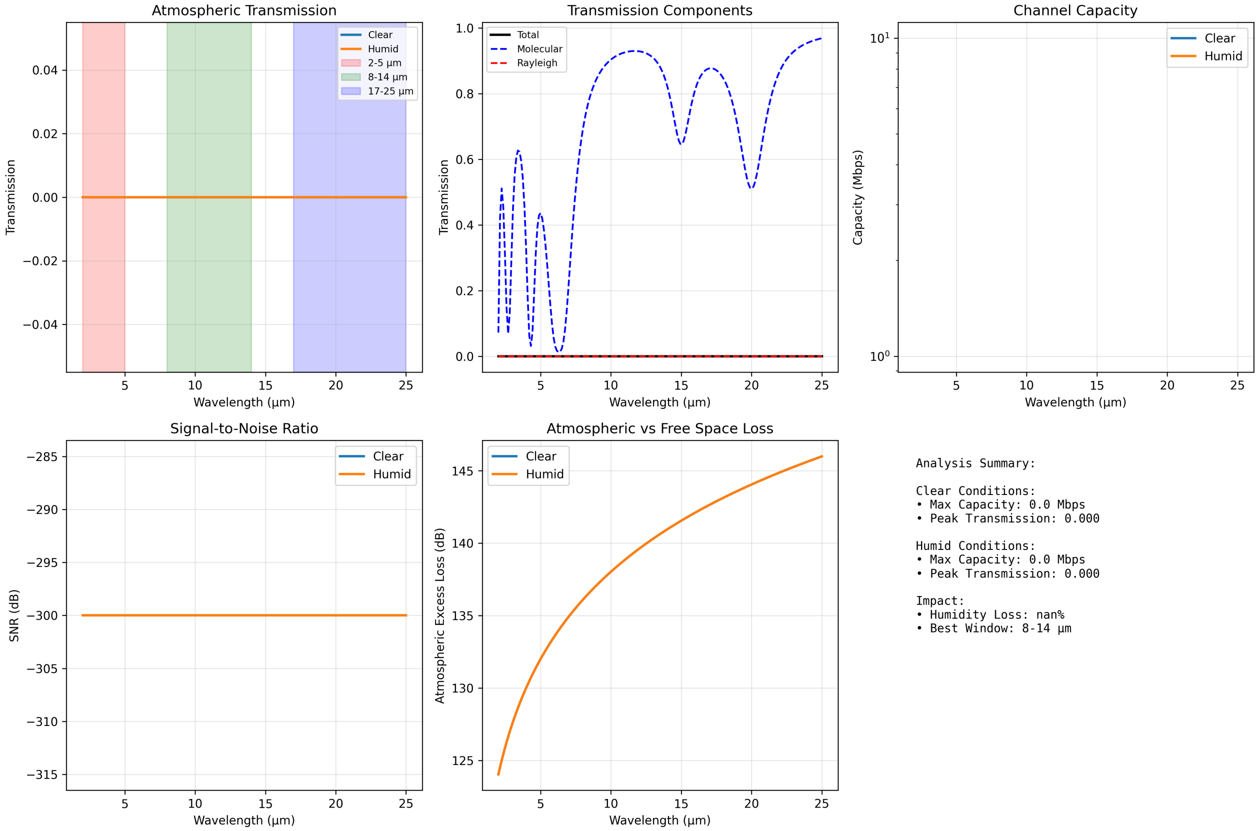

Figure 1: *Comprehensive environmental channel outputs produced deterministically by ‘scripts/generate$_e$nvironmental$_c$hannel$_a$nalysis.py‘using‘src/case$_s$tudies/environmental$_c$hannel.py‘.Panelsshowmolec grids, and optimized wavelength choices for target ranges.*

1.8 Cross-references

- Methods: sec:methodology
- Symbols: sec:symbols$_glossaryMathappendix : sec : mathematical_appendix$

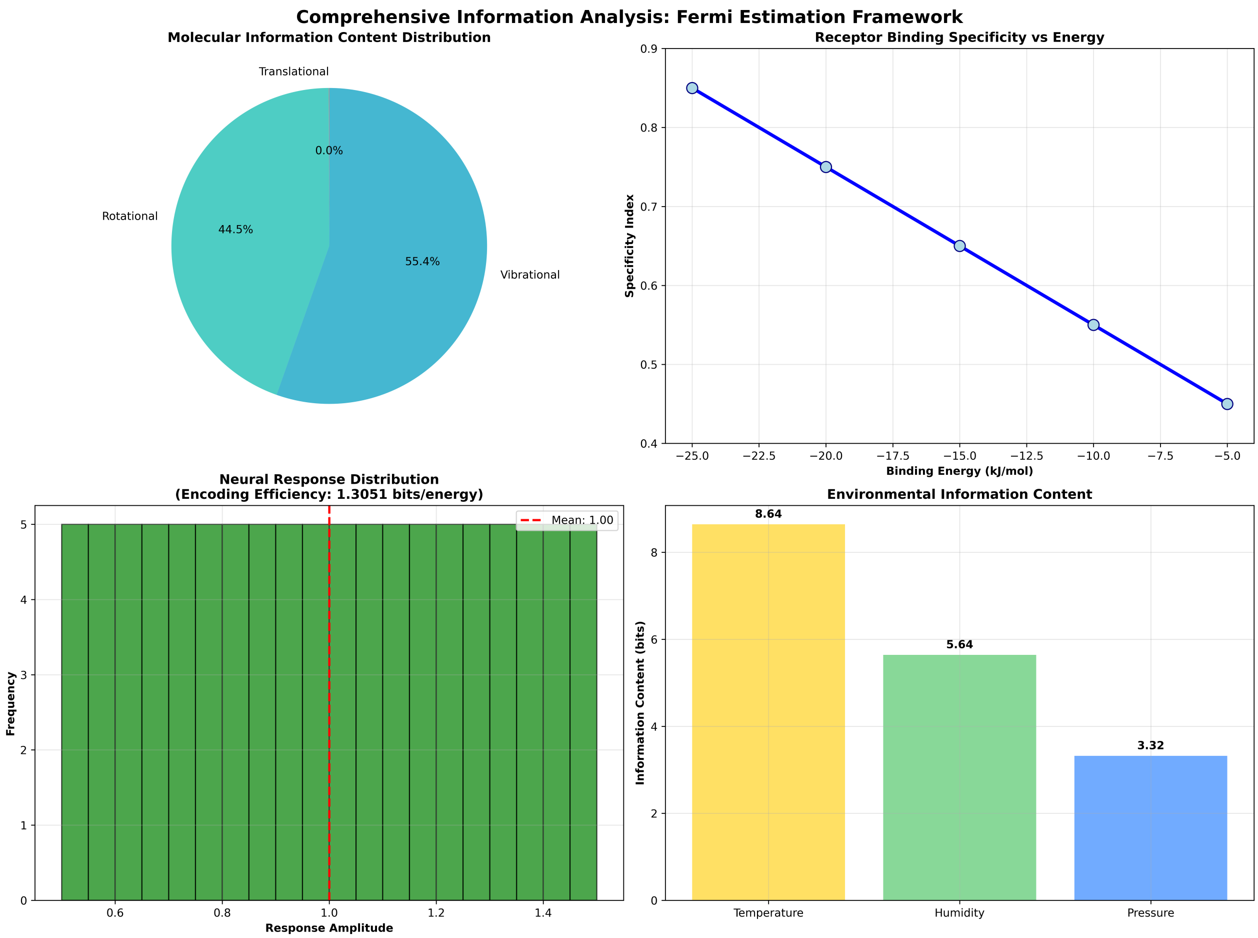

Figure 2: *Integrated information analysis (from ‘scripts/generate$_i$ntegrated$_a$nalysis.py‘)showingmolecular, receptor, neuralandenvironmentalinformationdecompositio*